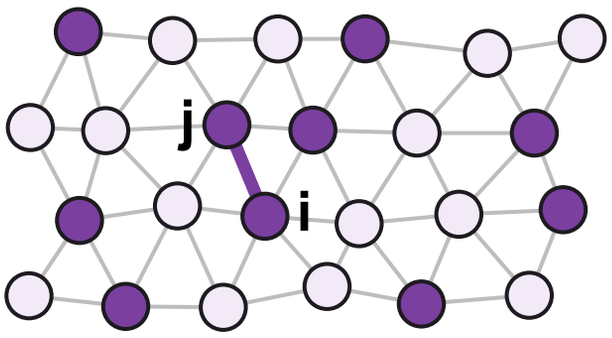
## a Labels on the neighbour graph




$x_i$ site fraction

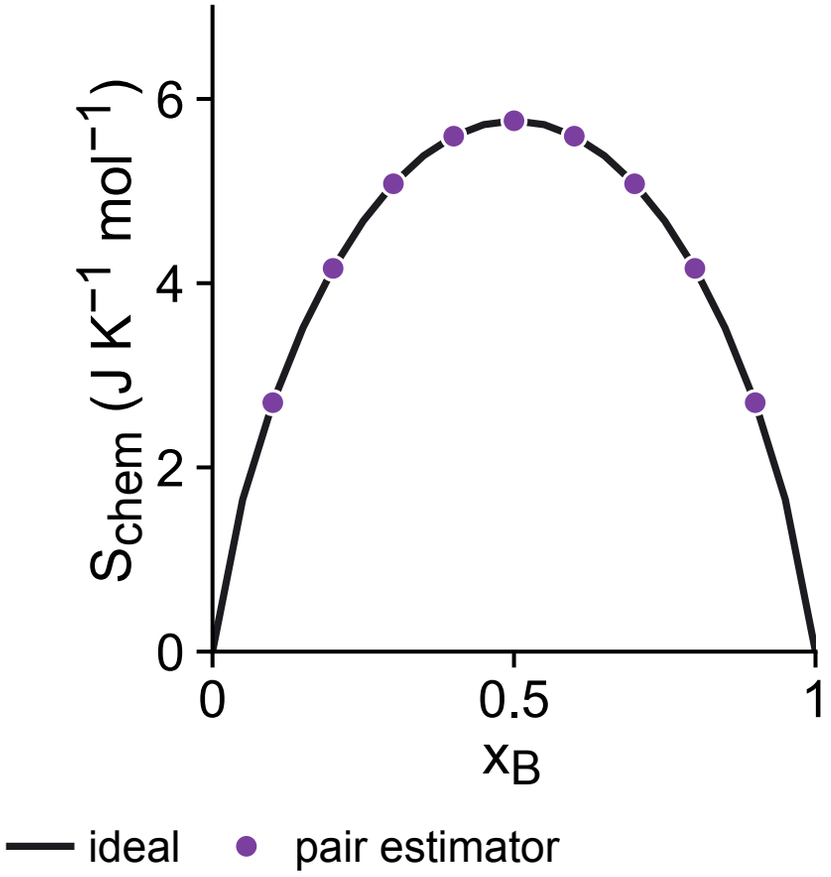
$y_{ij}$ bond fraction

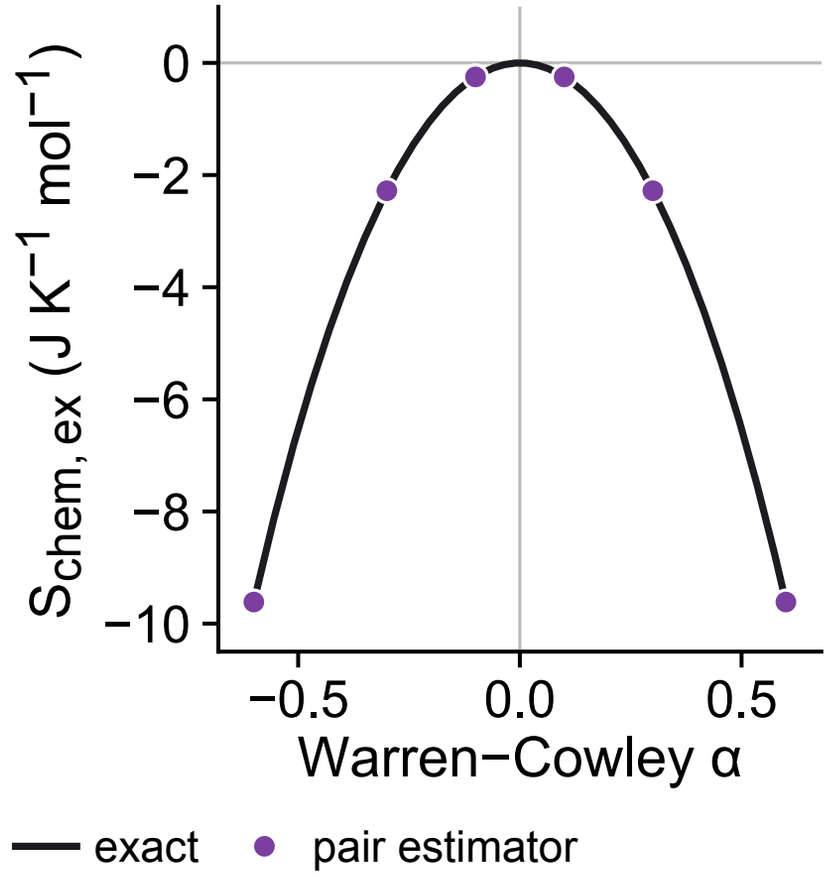
$w_i$ bond-end fraction

$\bar{z}$ mean neighbours

$$S_{chem}/R = -\sum_i x_i \ln x_i - (\bar{z}/2)\sum_{ij} y_{ij} \ln[y_{ij}/(w_i w_j)]$$

## b Ideal limit




## c Ordered benchmark